

Project Initialization and Planning Phase

Date	27 June 2026
Team ID	
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution)

The Human Development Index (HDI) Prediction System is designed to predict the Human Development Index of a country or region using machine learning techniques. HDI is an important indicator that measures the overall development of a country based on factors such as life expectancy, education, and income.

The proposed solution utilizes historical socio-economic datasets and machine learning algorithms to analyze patterns and accurately estimate HDI values. The system enables policymakers, researchers, and organizations to understand development trends and make informed decisions for sustainable growth.

Key features include automated data preprocessing, machine learning-based prediction, visualization of results, and performance evaluation using standard regression metrics.

Project Overview	
Objective	The primary objective of this project is to develop a machine learning model capable of accurately predicting the Human Development Index (HDI) using socio-economic indicators. The system aims to assist governments, researchers, and policy makers in understanding development trends and supporting data-driven decision making.
Scope	The project focuses on collecting and preprocessing HDI-related datasets, training regression models, evaluating prediction accuracy, and presenting the predicted HDI values through an easy-to-use web interface. The solution supports educational research and policy analysis by providing reliable development predictions.

Problem Statement	
Description	Accurately estimating the Human Development Index is challenging due to the complex relationships among economic, educational, and healthcare indicators. Manual analysis is time-consuming and often fails to capture hidden patterns within large datasets.
Impact	Inaccurate or delayed HDI estimation may lead to ineffective policy planning and inefficient allocation of resources. A machine learning-based prediction system helps improve forecasting accuracy, enabling better planning for education, healthcare, and economic development.
Proposed Solution	
Approach	Develop a machine learning-based prediction system that analyzes historical socio-economic indicators such as GDP per capita, life expectancy, expected years of schooling, mean years of schooling, and gross national income to predict the Human Development Index.
Key Features	• Data preprocessing and cleaning • Feature selection and normalization • Machine learning model training • HDI prediction using regression algorithms • Performance evaluation using MAE, RMSE and R² Score • Interactive web interface • Visualization of prediction results

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor	Intel Core i5 or higher
Memory	RAM specifications	8 GB
Storage	Disk space	20 GB Free SSD/HDD

Software		
Operating System	Platform	Windows 10/11 or Ubuntu
Programming Language	Development	Python 3.10+
Framework	Web Framework	Flask
Libraries	Machine Learning	Pandas, NumPy, Scikit-learn, Matplotlib, Joblib
IDE	Development Environment	VS Code
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Human Development Index Dataset (CSV) containing socio-economic indicators collected from public development datasets. Source: Kaggle Human Development Index Dataset